



Historical Data Testing

Results Report

Version 0.0.2 Beta

Testing Methodology

Testing was conducted with a slightly modified version of HedgeHog Version 0.0.2 Beta. The modifications include:

- Modifying all ratio calculation functions in “quant.py” to utilize data from previous dates.
- Modifying prompts in “quality.py” to analyze market sentiment from previous dates.

Two sets of tests were conducted utilizing data from one year ago, with March 17, 2025, serving as the reference date. HedgeHog was provided with 40 randomly selected companies from the S&P 500 and an imaginary budget of \$1,000 USD. The first set of tests was conducted with and the second without the qualitative analysis algorithm to mitigate any potential AI bias based on future knowledge. This approach also enables the assessment of the quantitative analysis algorithm’s effectiveness. The tests with and without AI were conducted on the same stocks for each test set.

Following HedgeHog’s decision-making process, the theoretical performance of its portfolio was compared with the S&P 500 SPX index, the Dow Jones total US stock market index, and the average performance of the full stock pool during the same period.

Stock pool

HedgeHog was given the following pool of stocks:

- | | | |
|----------|----------|----------|
| - 'MU' | - 'PEG' | - 'NDAQ' |
| - 'PSX' | - 'IRM' | - 'KEY' |
| - 'RCL' | - 'AIG' | - 'ROST' |
| - 'SRE' | - 'LULU' | - 'ED' |
| - 'POOL' | - 'NCLH' | - 'LKQ' |
| - 'LVS' | - 'EIX' | - 'MCD' |
| - 'CBRE' | - 'CRM' | - 'BALL' |
| - 'DVN' | - 'AMZN' | - 'DUK' |
| - 'FITB' | - 'K' | - 'WMB' |
| - 'QCOM' | - 'AON' | - 'FDS' |
| - 'DD' | - 'WBA' | - 'VTR' |
| - 'PHM' | - 'EL' | - 'AMGN' |
| - 'ZBH' | - 'EMR' | |
| - 'TER' | - 'PFE' | |

Test set 1

In this test set, given the stocks, HedgeHog's recommendations were the following:

- | | |
|------------------------------|------------------------------|
| - Buy 38.23\$ of VTR shares | - Buy 50.70\$ of DVN shares |
| - Buy 39.04\$ of ED shares | - Buy 51.15\$ of PFE shares |
| - Buy 39.30\$ of PSX shares | - Buy 51.94\$ of CRM shares |
| - Buy 41.84\$ of LKQ shares | - Buy 54.67\$ of SRE shares |
| - Buy 42.23\$ of LULU shares | - Buy 54.84\$ of NDAQ shares |
| - Buy 42.97\$ of EIX shares | - Buy 56.48\$ of ZBH shares |
| - Buy 43.08\$ of DUK shares | - Buy 56.83\$ of EMR shares |
| - Buy 44.18\$ of PEG shares | - Buy 57.58\$ of MU shares |
| - Buy 48.57\$ of CBRE shares | - Buy 62.18\$ of TER shares |
| - Buy 50.52\$ of DD shares | - Buy 73.68\$ of PHM shares |

The YTD ROI on HedgeHog's choices was quite low at 0.57%, compared to the 9.33% of the S&P 500 index and the 8.40% of the Dow Jones total US market index. The original stock pool managed to get a 2.91% YTD ROI. However, HedgeHog managed to perform rather well up until October 2024, with an 8.98% ROI on 17/12/2024. As a matter of fact, while HedgeHog's performance was slightly lower than the S&P 500 index and the Dow Jones total US market index, it outperformed the original stock pool every single month until October.

HedgeHog's portfolio appears to be relatively good at stopping losses in the first few months. In April 2024, when the entire market was down, HedgeHog's portfolio lost less than the S&P 500 index and the Dow Jones total US market index, and its losses were nearly half of those endured by the original stock pool. However, after October 2024, it suffers more losses than the S&P 500 index, the total market index and even the original stock pool.

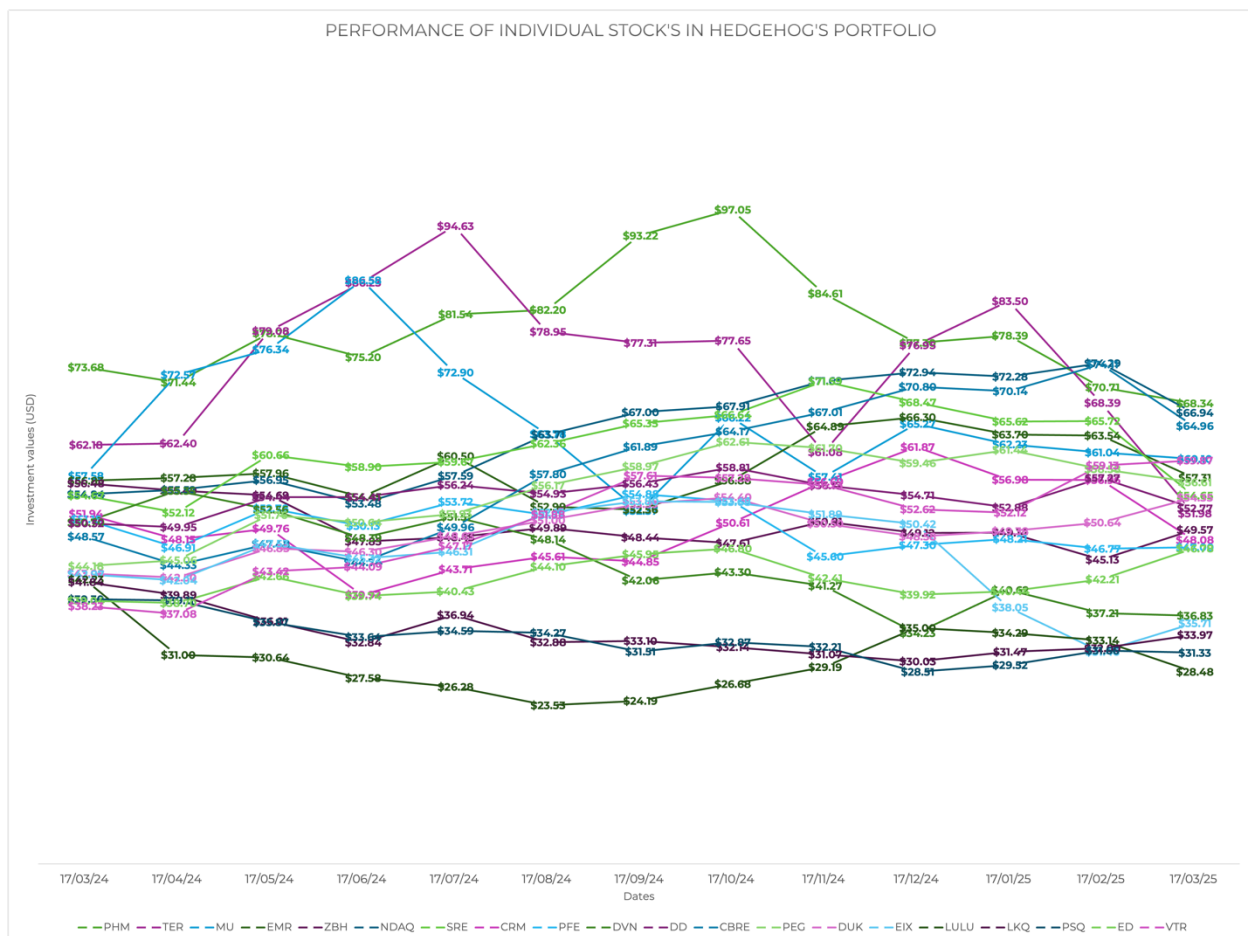


Chart 1: Performance of individual stocks in HedgeHog's set 1 portfolio

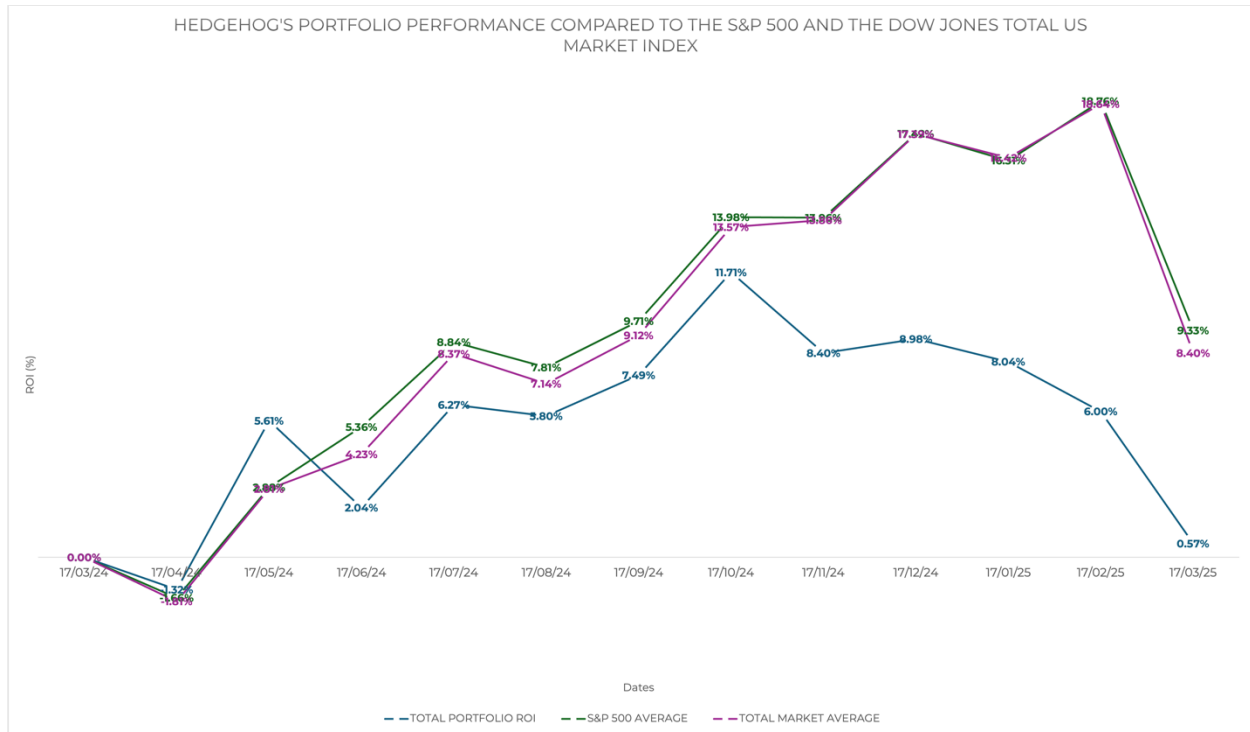


Chart 2: HedgeHog's set 1 portfolio performance compared to the S&P 500 and the Dow Jones total US market index



Chart 3: The performance of HedgeHog's set 1 portfolio compared to the original stock pool

The best-performing stocks in the original stock pool were the following:

Ticker	Annual ROI	Attributed score	Chosen?
RCL	65.69%	0.1348	No
VTR	54.67%	0.2887	Yes
WMB	54.41%	0.1156	No
CBRE	33.74%	0.3669	Yes
PEG	28.15%	0.3337	Yes
DUK	26.16%	0.3254	Yes
AON	24.23%	0.2712	No
NDAQ	22.07%	0.4142	Yes
ED	19.83%	0.2949	Yes
AMGN	16.53%	0.0896	No

Table 1: The best-performing stocks in the random stock pool, their ROIs and HedgeHog's attributed score for each company (set 1)

And the worst-performing stocks in the original stock pool were the following:

Ticker	Annual ROI	Attributed score	Chosen?
EL	-55.28%	0.0815	No
WBA	-45.74%	0.2483	No
LULU	-32.55%	0.3189	Yes
DVN	-27.35%	0.3830	Yes
PSX	-20.28%	0.2968	Yes
BALL	-19.44%	0.2780	No
LKQ	-18.19%	0.3160	Yes
POOL	-18.53%	0.1586	No
TER	-16.40%	0.4697	Yes
LVS	-14.71%	0.1033	No

Table 2: The worst-performing stocks in the random stock pool, their ROIs and HedgeHog's attributed score for each company (set 1)

Test set 2

In this test set, given the stock pool, HedgeHog's recommendations were the following:

- Buy 40.62\$ of AMZN shares
- Buy 40.62\$ of DUK shares
- Buy 41.32\$ of AMGN shares
- Buy 42.45\$ of SRE shares
- Buy 46.14\$ of VTR shares
- Buy 46.53\$ of EIX shares
- Buy 46.59\$ of TER shares
- Buy 47.81\$ of ED shares
- Buy 49.47\$ of CBRE shares
- Buy 50.67\$ of PSX shares
- Buy 51.46\$ of BALL shares
- Buy 51.80\$ of LKQ shares
- Buy 52.03\$ of CRM shares
- Buy 52.26\$ of PFE shares
- Buy 53.78\$ of EMR shares
- Buy 53.84\$ of LULU shares
- Buy 54.02\$ of DVN shares
- Buy 55.29\$ of MU shares
- Buy 57.01\$ of PHM shares
- Buy 66.31\$ of ZBH shares

The YTD ROI of HedgeHog's investment was negative, sitting at -2.02%. As a reminder, the S&P 500 index had an annual ROI of 9.33%, the Dow Jones total US market index sat at 8.40% and the original stock pool managed to get a 2.91% YTD ROI. However, the set 2 portfolio outperformed the set 1 portfolio in the first three months, where it achieved an ROI of 4.47%. It outperformed the original stock pool until July 2024, then fell behind. Similarly to the first set, the portfolio outperformed the S&P 500 and the total US market in the first three months. The second set's portfolio got less returns than the first set every single month.

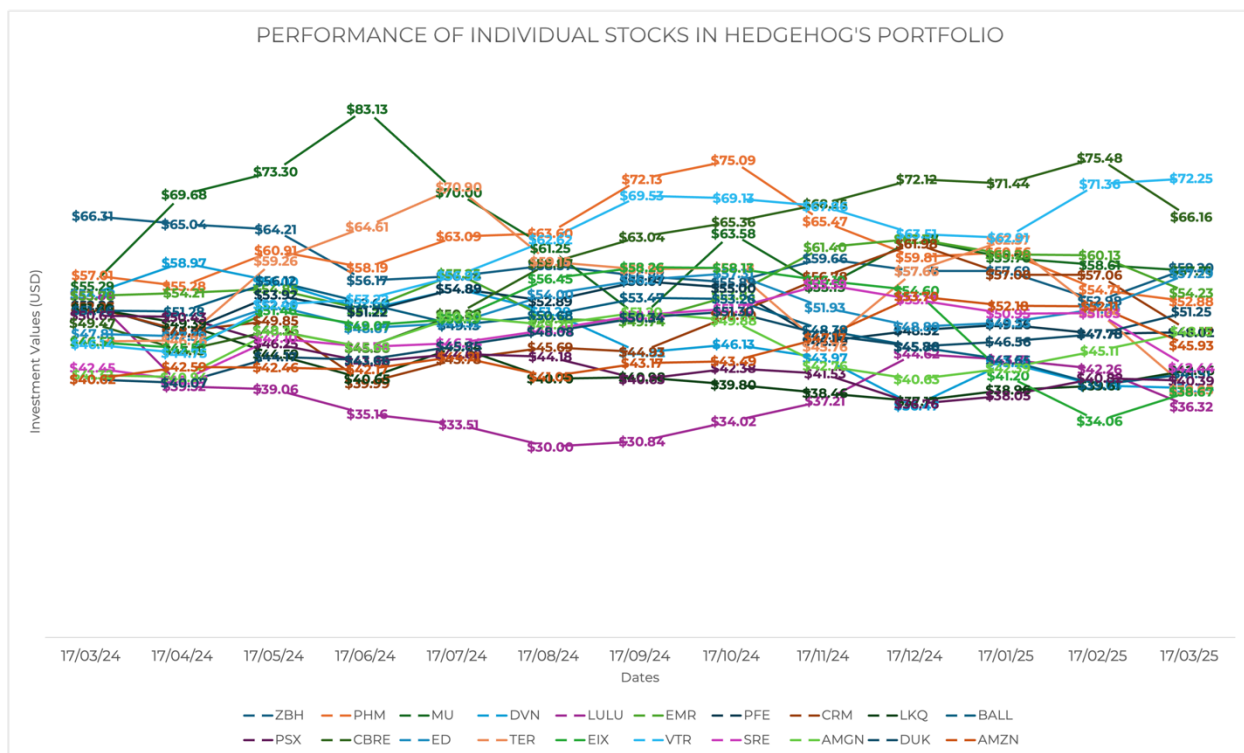


Chart 4: Performance of individual stocks in HedgeHog's set 2 portfolio

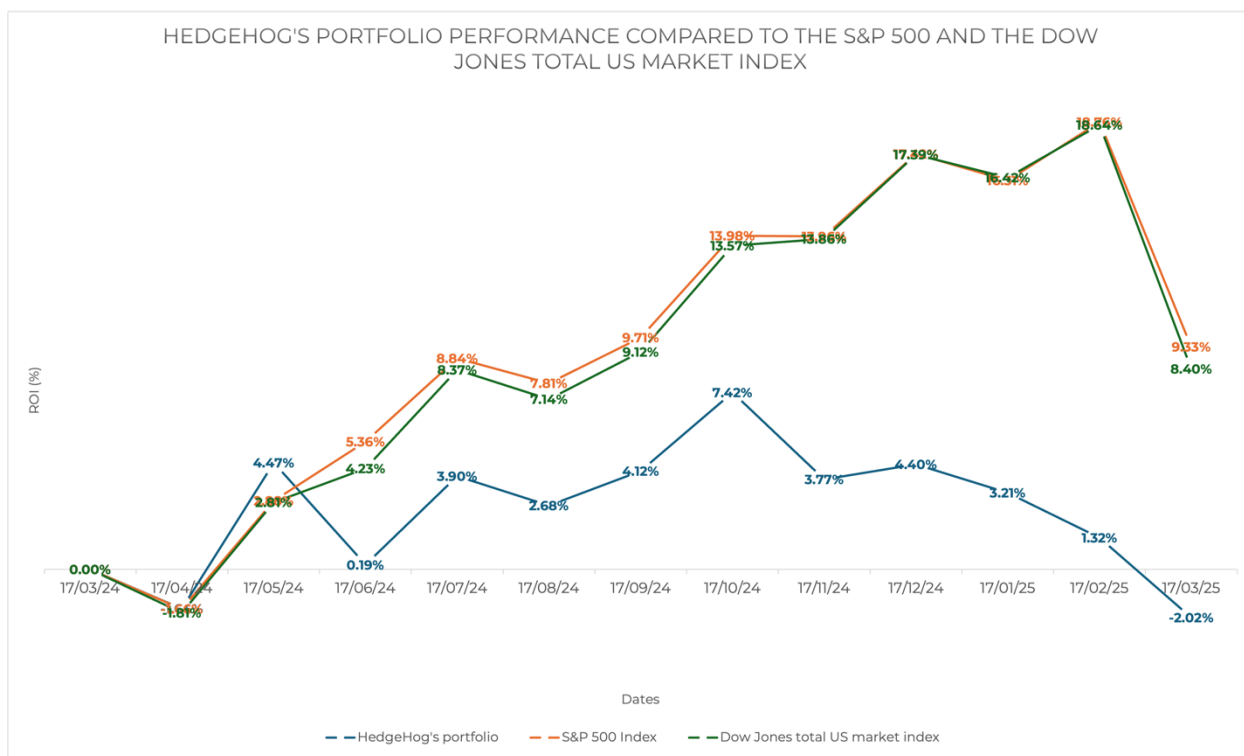


Chart 5: HedgeHog's set 2 portfolio performance compared to the S&P 500 and the Dow Jones total US market index

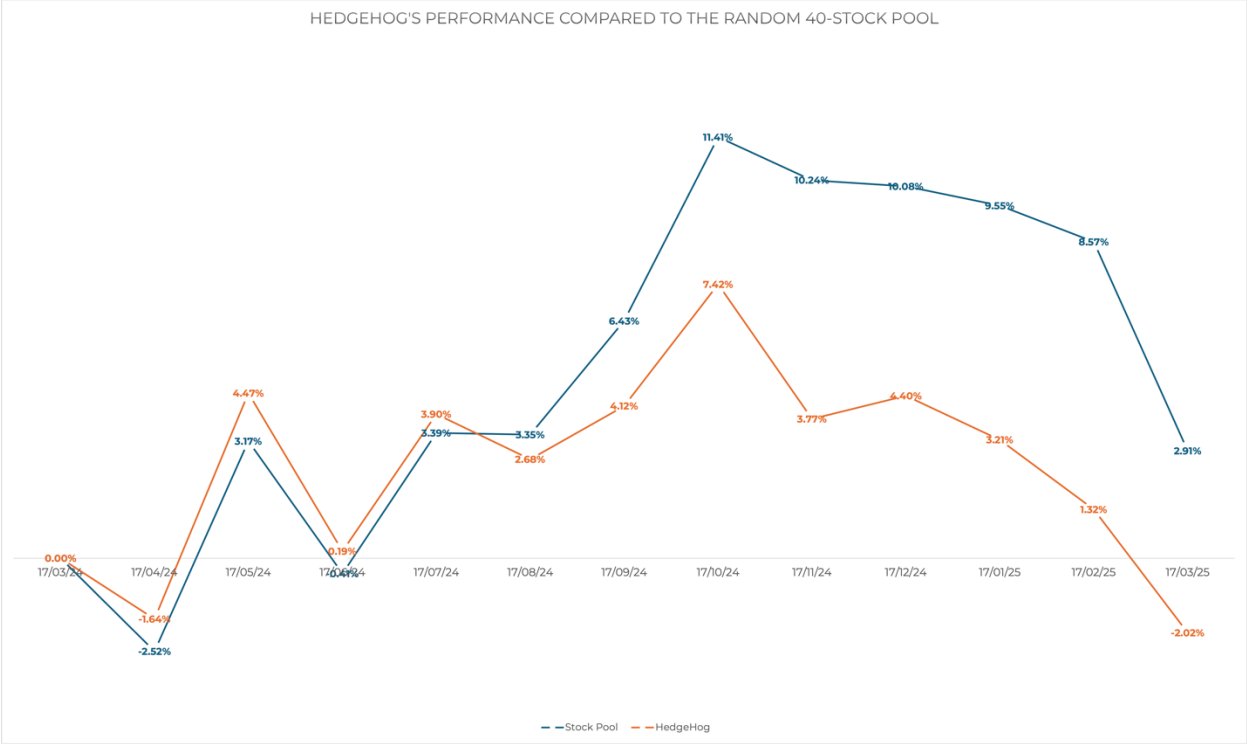
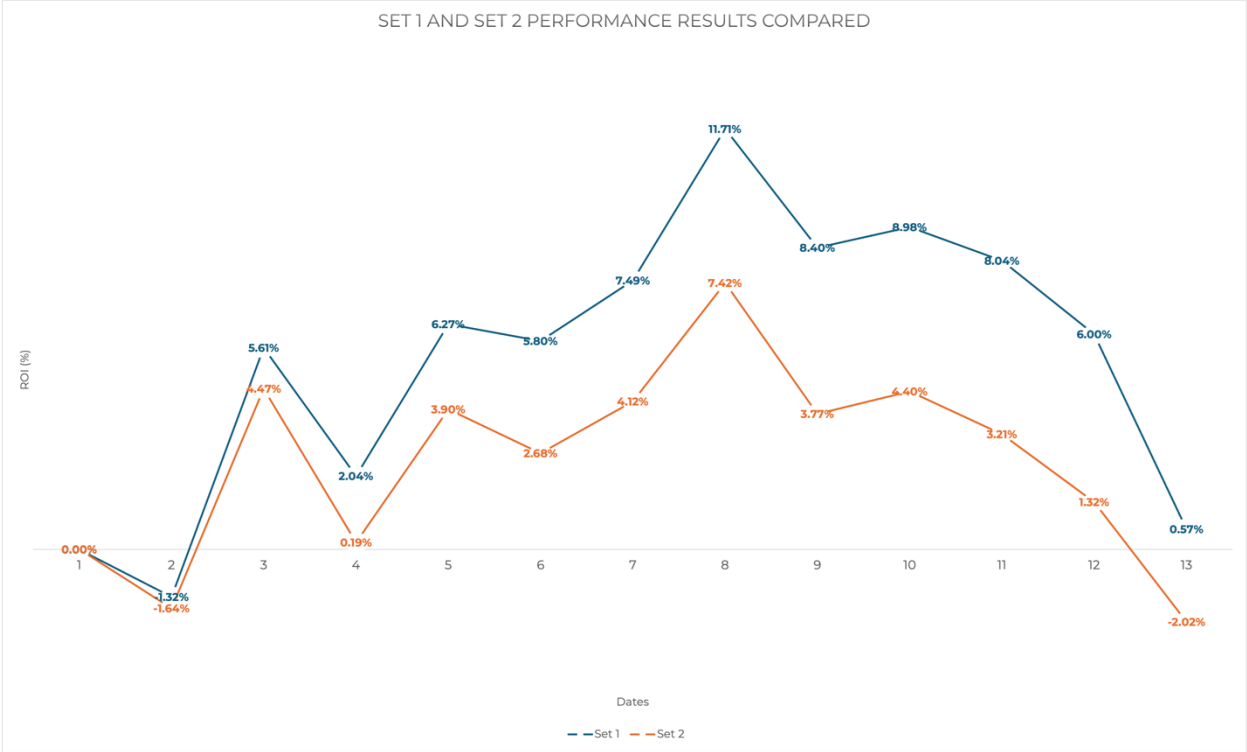


Chart 6: The performance of HedgeHog's set 2 portfolio compared to the original stock pool



Graph 7: Set 1 and Set 2 performance compared

The best-performing stocks in the original stock pool were the following:

Ticker	Annual ROI	Attributed score	Chosen?
RCL	65.69%	0.3031	No
VTR	54.67%	0.3820	Yes
WMB	54.41%	0.2165	No
CBRE	33.74%	0.4096	Yes
PEG	28.15%	0.3205	Yes
DUK	26.16%	0.3363	Yes
AON	24.23%	0.3055	No
NDAQ	22.07%	0.3240	No
ED	19.83%	0.3958	Yes
AMGN	16.53%	0.3420	No

Table 3: The best-performing stocks in the random stock pool, their ROIs and HedgeHog's attributed score for each company (set 2)

And the worst-performing stocks in the original stock pool were the following:

Ticker	Annual ROI	Attributed score	Chosen?
EL	-55.28%	0.2911	No
WBA	-45.74%	0.2927	No
LULU	-32.55%	0.4457	Yes
DVN	-27.35%	0.4472	Yes
PSX	-20.28%	0.4195	Yes
BALL	-19.44%	0.4260	Yes
LKQ	-18.19%	0.4288	Yes
POOL	-18.53%	0.3005	No
TER	-16.40%	0.3857	Yes
LVS	-14.71%	0.2561	No

Table 4: The worst-performing stocks in the random stock pool, their ROIs and HedgeHog's attributed score for each company (set 2)

Discussion

These tests show that the performance of HedgeHog's quantitative analysis algorithm is relatively decent, although there are areas that can be improved upon. The algorithm seems to perform best for medium-term investments of 3-6 months. It should be noted that our reference date being set in March means that the algorithm's information regarding the company's financial well-being is relatively recent, but as time goes on, the performance of its portfolio starts becoming more unpredictable. The use of quarterly information when available may help in improving performance over a long period.

The choice of financial parameters in the quantitative analysis algorithm appears to be good but insufficient. High-return investments tend to receive a good score, yet not all companies who receive good scores perform well. This may indicate that there are other important factors that are unaccounted for.

The AI-powered market sentiment analysis algorithm, however, seems to be severely flawed. The scores given to companies in the second test set are much less correlated with the company's performance, and the overall portfolio ROI is drastically lower. During testing, it was noticed that the AI usually tends to give a 0.6 or 0.7 score to all companies in all four criteria, with only a few rare exceptions. This poor performance means that to continue development, the qualitative analysis algorithm needs major revisions, including a possible removal of AI.

As a final note, one cannot help but notice that the sudden and dramatic drop in performance present in both of HedgeHog's portfolios in November 2024 coincides with the release of the 2024 United States presidential election results. There is no indication that this fact influenced the performance of HedgeHog's portfolio, but it may have had indirect effects on market sentiment towards the companies HedgeHog had chosen.